Simulation methods in Medical Engineering
Spring 2025
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CM2014 Mathematical and numerical background

Lecture notes for lecture 2

Analytical and numerical stability for systems

Analysis of the vibration equation

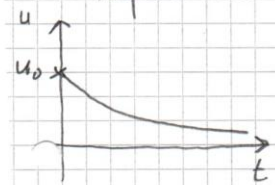
Overview of Lab 1

Simulation methods in Medical Engineering

HL2008, Mathematical and numerical background

Notes to lecture 2.

Repetition: MP1, $\frac{du}{dt} = \lambda u$, $u(0) = u_0$



Solution $u(t) = u_0 e^{\lambda t}$ stable if $Re(\lambda) \leq 0$

Here we call this analytical stability.

1) Explicit Euler method applied to MP1:

$$\begin{cases} u_k = u_{k-1} + h\lambda u_{k-1} \\ t_k = t_{k-1} + h \end{cases} \quad k = 1, 2, \dots, N$$

For a general nonlinear scalar ODE

$$\frac{du}{dt} = f(t,u), \quad u(0) = u_0 \qquad (*)$$

$$\begin{cases} u_k = u_{k-1} + h f(t_{k-1}, u_{k-1}) \\ t_k = t_{k-1} + h \end{cases}, \quad k = 1, 2, \dots, N$$

2) Implicit Euler method applied to MP1:

$$\begin{cases} u_k = u_{k-1} + h\lambda u_k \Rightarrow u_k = \frac{1}{1-h\lambda} u_{k-1} \\ t_k = t_{k-1} + h \end{cases}$$

Applied to $(*)$

$$\begin{aligned} u_k &= u_{k-1} + h f(t_k, u_k) \leftarrow \text{nonlinear} \\ t_k &= t_{k-1} + h \end{aligned}$$

algebraic eqn to be solved w.r.t. u_k

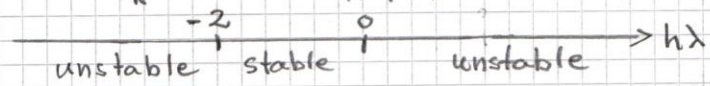
Both methods have 1st order accuracy

i.e. $u(t_k) - u_k = O(h)$

Numerical stability for EE and IE:

1) For EE applied to MP1

$u_k = (1+h\lambda)u_{k-1}$, stable if $|1+h\lambda| \leq 1$

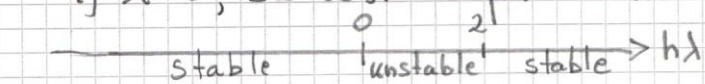


if $\lambda < 0$, h must fulfill $0 < h \leq h_s = \frac{2}{-\lambda}$

2) For IE applied to MP1

$u_k = \frac{1}{1-h\lambda} u_{k-1}$, stable if $|1-h\lambda| \geq 1$

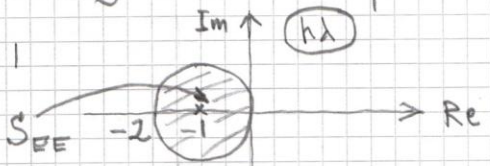
if $\lambda < 0$, IE is stable for all $h > 0$



What about stability when λ complex ?

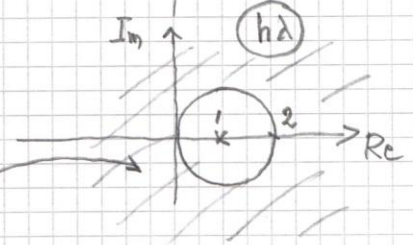
EE: $|1+h\lambda| \leq 1$

S_{EE} stability region for EE



IE: $|1-h\lambda| \geq 1$

S_{IE} stability region for IE



Note: In some textbooks stability is referred to as absolute stability

Best would be if the stability region for the exact solution ($Re(\lambda) \leq 0$) is the same as the stability region for the method ($Re(h\lambda) \leq 0$)

Such a method exists: the trapezoidal method (LE 3.3.5)

$$\begin{cases} u_k = u_{k-1} + \frac{h}{2}\left(f(t_{k-1}, u_{k-1}) + f(t_k, u_k)\right) \\ t_k = t_{k-1} + h \end{cases}$$

The method is implicit (u_k at both side) and second order accurate: $u(t_k) - u_k = O(h^2)$

However there are disadvantages with oscillations in the numerical solution u_k, see trapets.m

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What can be said about stability for a linear system of ODEs:

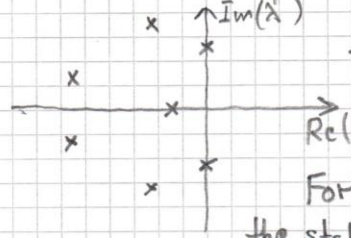
$$\frac{d\bar{u}}{dt} = A\bar{u}\,, \quad \bar{u}(0) = \bar{u}_0 \qquad (*)$$

where $\bar{u}, \bar{u}_0$ are $n \times 1$ vectors
A is an $n \times n$ matrix

Without proof we state:
The system (*) is <u>analytically stable</u> if

$$\mathrm{Re}(\lambda_i) \leq 0\,, \quad i = 1, 2, \ldots, n$$

where λ_i are the <u>eigenvalues</u> of A.
(If $\mathrm{Re}(\lambda_i) = 0$ we assume that λ_i is <u>simple</u>, i.e. no multiple eigenvalues)



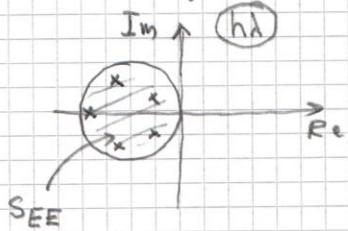
← A system with these eigenvalues is stable.

For nonlinear systems $\dot{\bar{u}} = \bar{f}(t, \bar{u})$

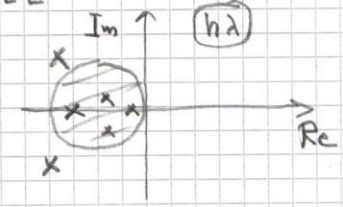
What about numerical stability?

EE: $\bar{u}_k = \bar{u}_{k-1} + hA\bar{u}_{k-1}$, $k = 1, 2, \dots, N$
$t_k = t_{k-1} + h$

Stable if $h\lambda_i \in S_{EE}$, $i = 1, 2, \dots, n$

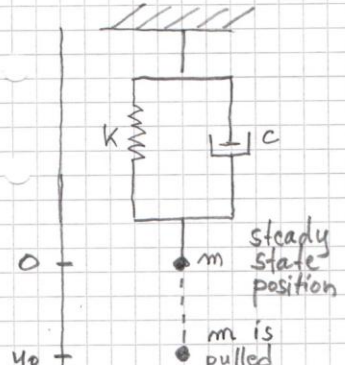


This positioning of $h\lambda_i$ will be stable

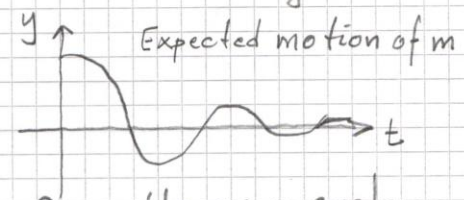


This positioning of $h\lambda_i$ will be unstable

Example: Damped mechanical vibrations
(LE Ex 1.1 + Fig 1.4, pg 6-7)



Vibration eqn:
$m\ddot{y} + c\dot{y} + ky = 0$, $y(0) = y_0$
$\dot{y}(0) = 0$



1) Rewrite as a system

$u_1 = y$, $u_2 = \dot{y}$

$$\begin{cases} \dot{u}_1 = u_2 & u_1(0) = y_0 \\ \dot{u}_2 = -\frac{c}{m}u_2 - \frac{k}{m}u_1 & u_2(0) = 0 \end{cases}$$

component form

$$\dot{\bar{u}} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix}\bar{u}, \quad \bar{u}_0 = \begin{pmatrix} y_0 \\ 0 \end{pmatrix}$$ ← matrix form

u_1, u_2 dependent variables
t independent variable
m, c, k, y_0 parameters

Explit Euler applied to the component form:

$$u_{1,k} = u_{1,k-1} + h u_{2,k-1}, \quad u_{1,0} = y_0$$
$$u_{2,k} = u_{2,k-1} + h\left(-\frac{c}{m} u_{2,k-1} - \frac{k}{m} u_{1,k-1}\right), \quad u_{2,0} = 0$$

Let $m = 1$, $c = 0.4$, $k = 4.5$, $y_0 = 1$ $(= u_{1,0})$

$h = 0.1$

$u_{1,1} = u_{1,0} + h u_{2,0} = 1 + 0.1 \cdot 0 = 1$

$u_{2,1} = u_{2,0} + h(-0.4\, u_{2,0} - 4.5\, u_{1,0}) = -0.45$

$t_1 = t_0 + h = 0 + 0.1 = 0.1$

$u_{1,2} =$

$u_{2,2} =$

$t_2 =$

The system on matrix form with EE:

$$\dot{\bar{u}} = A\bar{u}, \ \bar{u}(0) = \bar{u}_0, \quad A = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -4.5 & -0.4 \end{pmatrix}, \ \bar{u}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

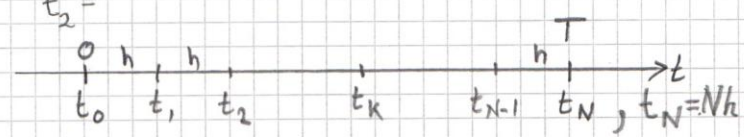
$$\bar{u}_k = \bar{u}_{k-1} + hA\bar{u}_{k-1}$$

$$\bar{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0.1 \begin{pmatrix} 0 & 1 \\ -4.5 & -0.4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -0.45 \end{pmatrix} = \begin{pmatrix} 1 \\ -0.45 \end{pmatrix}$$

$t_1 = t_0 + h = 0 + 0.1 = 0.1$

$\bar{u}_2 =$

$t_2 =$



Explit Euler applied to the component form:

$$u_{1,k} = u_{1,k-1} + h u_{2,k-1}, \quad u_{1,0} = y_0$$
$$u_{2,k} = u_{2,k-1} + h\left(-\frac{c}{m} u_{2,k-1} - \frac{k}{m} u_{1,k-1}\right), \quad u_{2,0} = 0$$

Let $m = 1$, $c = 0.4$, $k = 4.5$, $y_0 = 1$ $(= u_{1,0})$

$h = 0.1$

$k=1$:
$$\begin{cases} u_{1,1} = u_{1,0} + h u_{2,0} = 1 + 0.1 \cdot 0 = 1 \\ u_{2,1} = u_{2,0} + h(-0.4\, u_{2,0} - 4.5\, u_{1,0}) = -0.45 \\ t_1 = t_0 + h = 0 + 0.1 = 0.1 \end{cases}$$

$k=2$:
$$\begin{cases} u_{1,2} = u_{1,1} + h u_{2,1} = 1 - 0.1 \cdot 0.45 = 0.955 \\ u_{2,2} = u_{2,1} + h(-0.4 \cdot 0.45 - 4.5 \cdot 1) = -0.882 \\ t_2 = t_1 + h = 0.1 + 0.1 = 0.2 \end{cases}$$

The system on metrix form with EE:

$$\dot{\bar{u}} = A\bar{u}, \ \bar{u}(0) = \bar{u}_0, \quad A = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -4.5 & -0.4 \end{pmatrix}, \ \bar{u}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

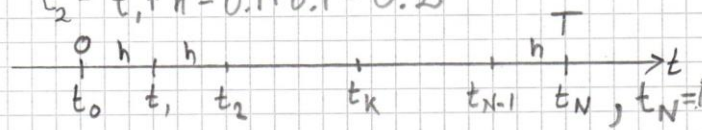
$$\bar{u}_k = \bar{u}_{k-1} + hA\bar{u}_{k-1}$$

$$\bar{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0.1 \begin{pmatrix} 0 & 1 \\ -4.5 & -0.4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -0.45 \end{pmatrix} = \begin{pmatrix} 1 \\ -0.45 \end{pmatrix}$$

$$t_1 = t_0 + h = 0 + 0.1 = 0.1$$

$$\bar{u}_2 = \begin{pmatrix} 1 \\ -0.45 \end{pmatrix} + 0.1 \begin{pmatrix} 0 & 1 \\ -4.5 & -0.4 \end{pmatrix} \begin{pmatrix} 1 \\ -0.45 \end{pmatrix} = \begin{pmatrix} 0.955 \\ -0.882 \end{pmatrix}$$

$$t_2 = t_1 + h = 0.1 + 0.1 = 0.2$$



Written on matrix form

$$\dot{\bar{u}} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix} \bar{u}, \quad \bar{u}(0) = \begin{pmatrix} y_0 \\ 0 \end{pmatrix}, \quad t \in [0, t_N]$$

Let $m=1$, $c=0.4$, $k=4.5$, $y_0 = 1$

MATLAB-program with EE: exeuldamp.m

$h=0.1$ gives an unstable solution, why?

$$A = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -4.5 & -0.4 \end{pmatrix}$$

Eigenvalues λ_1, λ_2 obtained from

$$\det(A - \lambda I) = 0$$

$$\det \begin{pmatrix} -\lambda & 1 \\ -4.5 & -0.4-\lambda \end{pmatrix} = \lambda(0.4+\lambda) + 4.5 = 0$$

$$\lambda_1 = -0.2 + 2.1119\,i, \quad \lambda_2 = -0.2 - 2.1119\,i$$

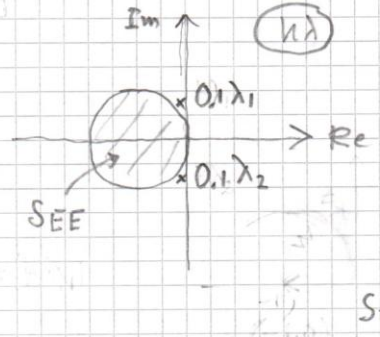
Alternatively obtained in MATLAB from

eig(A) giving λ_1 and λ_2 above

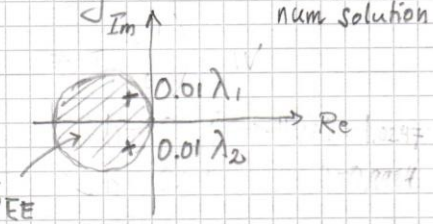
$\mathrm{Re}(\lambda_i) = -0.2 < 0 \Rightarrow$ analytically stable

BUT with $h=0.1$

$\Rightarrow h\lambda_1 = -0.02 + 0.21119\,i$, $h\lambda_2 = -0.02 - 0.21119\,i$ are situated outside S_{EE} $\Rightarrow$ unstable numerical solution



Decrease h to say $h=0.01$: stable num solution



MATLAB-program with IE: imeuldamp.m

$$\bar{u}_k = \bar{u}_{k-1} + hA\bar{u}_k \qquad \bar{u}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$(I - hA)\bar{u}_k = \bar{u}_{k-1}$$ Linear system of eqns.

$$\bar{u}_k = (I - hA) \backslash \bar{u}_{k-1}$$ Gaussian elimination to solve the eqns

↑ MATLAB-notation

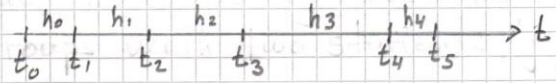
$$\begin{pmatrix} 1 & -0.1 \\ 0.45 & 1.04 \end{pmatrix} \bar{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow \bar{u}_1 = \begin{pmatrix} 0.96 \\ -0.41 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -0.1 \\ 0.45 & 1.04 \end{pmatrix} \bar{u}_2 = \begin{pmatrix} 0.96 \\ -0.41 \end{pmatrix} \Rightarrow \bar{u}_2 = \begin{pmatrix} 0.88 \\ -0.78 \end{pmatrix}$$

Use MATLABs built in ODE-solver ode45

Program damp.m + function fdamp.m

In ode45 automatic stepsize control is used giving variable stepsize (LE pg 59) $h_0, h_1, h_2, \ldots h_N$. In most cases this is more efficient than constant stepsize h.



The numerical method used in ode45 also has higher order accuracy than EE and IE.

Overview of LAB1

To sum up:

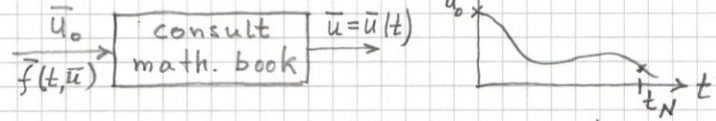
Numerical solution of IVP

1, Formulate in system form

$$\frac{d\bar{u}}{dt} = \bar{f}(t,\bar{u}), \quad \bar{u}(0) = \bar{u}_0$$

2, Solution by

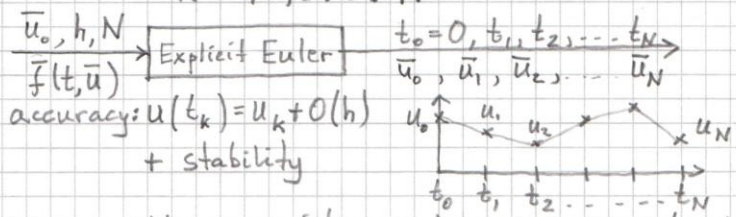
a, Analytical method



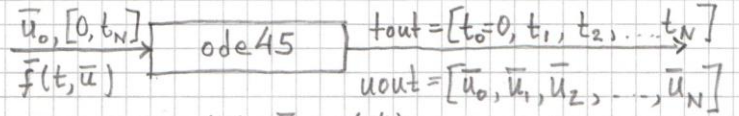
b, Numerical method from scratch

Ex. Explicit Euler, constant h

$$\bar{u}_k = \bar{u}_{k-1} + h\bar{f}(t_{k-1}, \bar{u}_{k-1}), \quad \bar{u}_0 \text{ given}$$
$$t_k = t_{k-1} + h, \quad t_0 = 0$$
$$k = 1, 2, \ldots N$$



c, Use an existing code, e.g. MATLAB's ode45



accuracy: $u(t_k) = \bar{u}_k + O(h^4)$